

Relational Model & SQL Basics

1. Relational Model Concepts

The relational model represents data as a collection of tables. A table is called a relation. Each row in a table represents a collection of related data values. These rows are called tuples. The columns are called attributes.

2. Types of Keys

- Super Key: A set of one or more attributes that can uniquely identify a tuple.
- Candidate Key: A minimal super key; a super key with no redundant attributes.
- Primary Key: A candidate key chosen by the database designer as the principal means of identifying entities.
- Foreign Key: An attribute or set of attributes in one relation that refers to the primary key in another relation.

3. SQL DDL (Data Definition Language)

DDL is used to define database structures or schemas.

- CREATE: To create objects in the database (e.g., tables).
- ALTER: To alter the structure of the database.
- DROP: To delete objects from the database.

4. SQL DML (Data Manipulation Language)

DML is used for managing data within schema objects.

- SELECT: Retrieve data from the database.
- INSERT: Insert data into a table.
- UPDATE: Update existing data within a table.
- DELETE: Delete records from a table.

5. Filtering and Grouping

The WHERE clause is used to filter records based on specific conditions. The ORDER BY clause is used to sort the result-set in ascending or descending order.

The GROUP BY statement groups rows that have the same values into summary rows. The HAVING clause was added to SQL because the WHERE keyword cannot be used with aggregate functions.